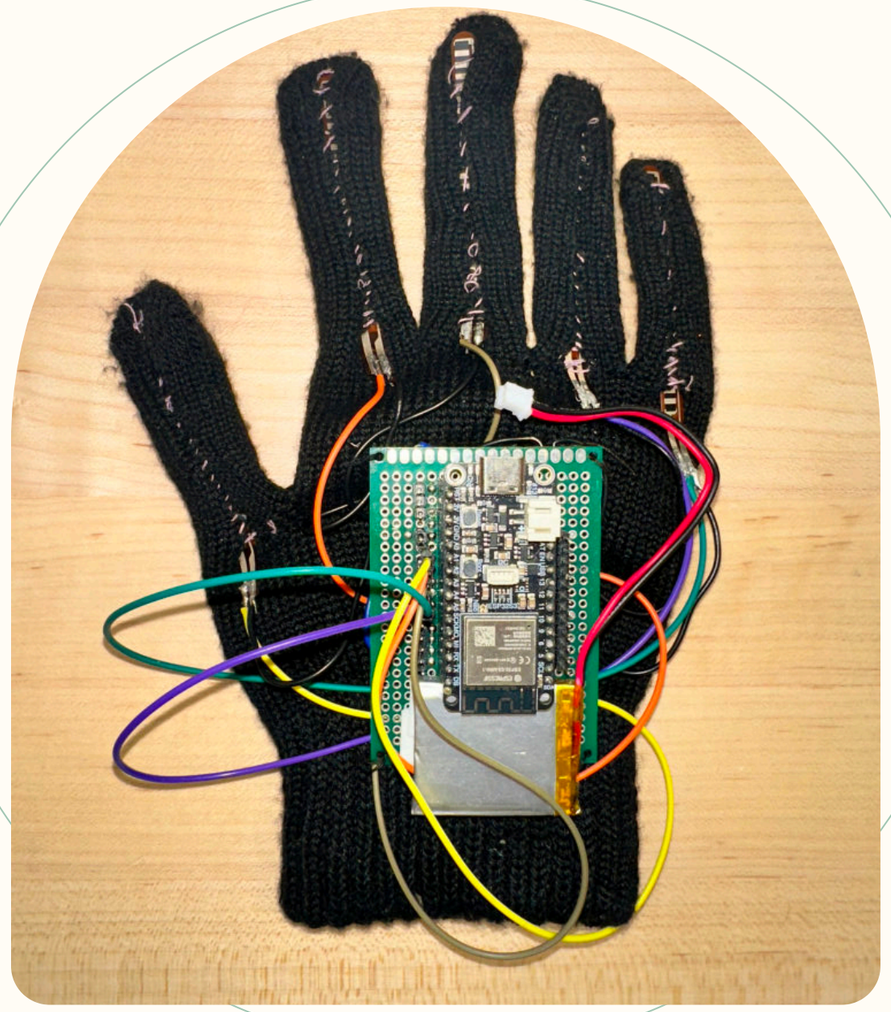
Helping Hand

ASL practice.
Feedback from your own hand.

A reference shows the sign.
Our goal is feedback on your attempt.

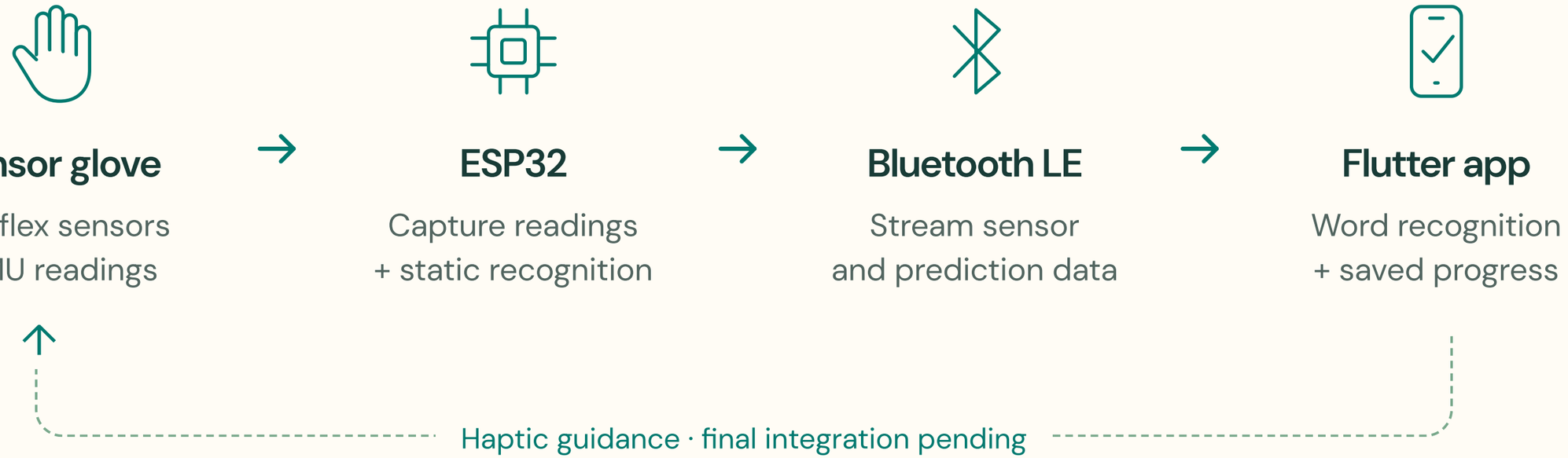
Srinitha Srikanth
Brian Paz
Gael Garcia
Kali Schuchhardt



October beta glove · final photo to replace

Wear. Sign. Get feedback.

Finger bending and motion become a practice response.



Static poses → MLP on the glove

Classify a held hand configuration.

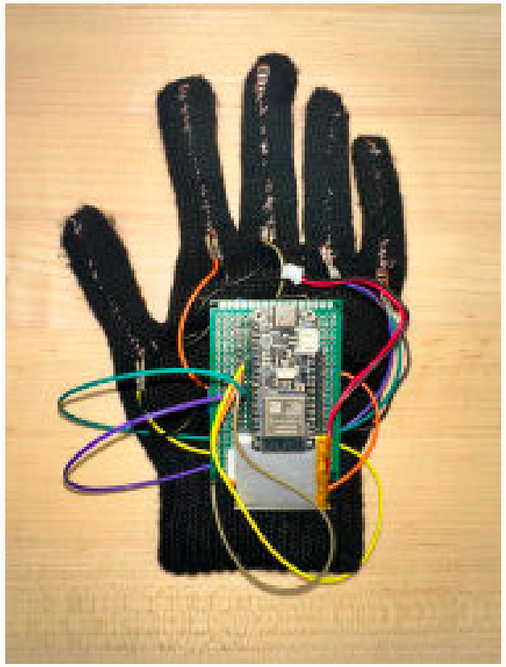
Motion sequences → TCN on the phone

Classify a complete word attempt.

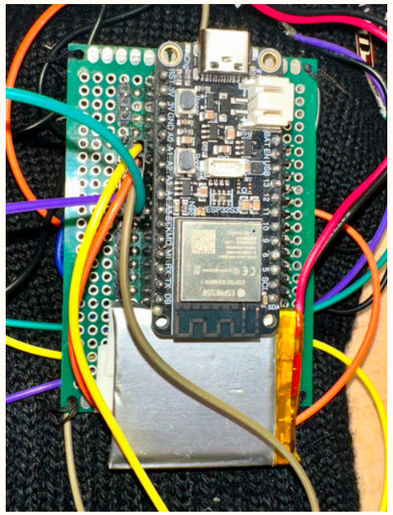
Current recognition locations · dashed return path is planned

A more integrated glove.

Final-build goals: PCB, new IMU, and feedback you can feel.



October glove



October perfboard

A board made for the glove

Custom PCB and final mounting.

ASSEMBLY PHOTO PENDING

A new view of motion

ICM-20948 accelerometer, gyro, magnetometer.

ON-GLOVE RESULTS PENDING

Guidance at the hand

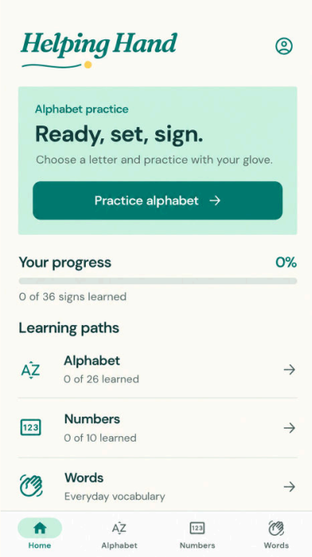
Planned: five finger + five dorsal piezos.

FINAL CUE BEHAVIOR PENDING

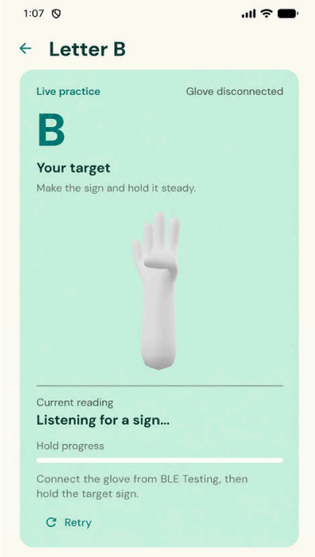
Final assembly + PCB close-up

Photo spaces reserved for the November build

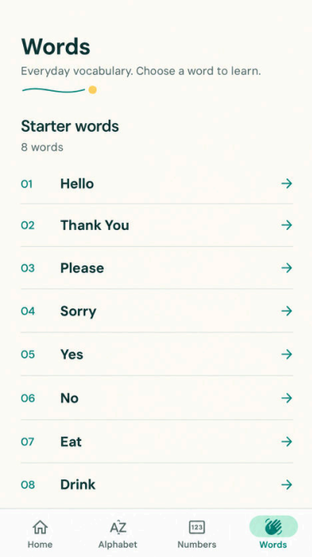
One place to practice.



Choose



Practice



Explore words

October UI references · practice screen is disconnected.
Replace with final connected-practice and completion screens.

November 19 recognition target

26	10	8
letters	digits	words

Hello trained	Thank You
Please trained	Sorry
Yes trained	No
Eat	Drink

> Current: 34 static classes + 3 trained words.
> Eight-word expansion and motion letters J/Z pending.

Practice loop

Select → attempt → feedback → save

Final measurements

[Add fresh results, trial scope, and phone latency]

Targets are goals · “trained” identifies the current three-word bundle

Try it.
Feel it.
Learn the sign.



Sign → Feedback → Progress

Live glove + phone · 90 seconds
Final targets and haptic sequence to rehearse.

Live use · show the actual response and outcome

Three pieces that came together.

01

A connected practice loop

CHALLENGE

Make sensor data useful to a learner.

APPROACH

Connect BLE, feedback, and saved progress.

OUTCOME

Historical live practice and saved completion reported.

OCTOBER TEAM OBSERVATION

02

From poses to motion

CHALLENGE

A word changes over time.

APPROACH

Record attempts, compare models, integrate a TCN.

OUTCOME

Three-word inference and completion integrated.

SOFTWARE + HISTORICAL DEMO

03

A more integrated wearable

CHALLENGE

Bring electronics and feedback onto the glove.

APPROACH

PCB, mounting, IMU, and haptic integration.

OUTCOME

[Add final build outcome and why it worked]

CANDIDATE · FINAL BUILD PENDING

> Confirm three distinct stories and the team's explanation of why each approach worked.

The hard parts. The lessons.

DIFFICULTY + IMPACT	WHAT WE WOULD DO DIFFERENTLY
01 **Sensor faults + calibration** Bad readings complicate recognition and make old results harder to reuse.	Check signals early and often. Track mounting, calibration, and changes.
02 **Learning motion from limited data** Few trials, similar signs, and variable timing limit what a model result can establish.	Plan data coverage with the features. Use fresh attempts and include failure cases.
03 **Physical + software dependencies** Changes to the glove affect the full loop. [Add one concrete team example]	Integrate smaller pieces sooner. Agree interfaces and leave time for live use.

> Add what actually happened, why it was difficult, and how the team responded.

Bring the feedback to the learner.

The glove, recognition, and app are one experience.

Questions?

Working takeaway: test the whole loop.
[Add final demonstrated outcome + remaining limit]